

Jonathon Hirschi

Department of Mathematical & Statistical Sciences
University of Colorado Denver
jhirschi206@gmail.com
github.com/jh-206

Research Interests

Time series modeling and machine learning, with emphasis on recurrent neural networks; spatial statistics; and environmental applications, particularly wildfire modeling and fuel moisture prediction.

Education

PhD, Applied Mathematics. University of Colorado Denver May 2026

- Advisor: Jan Mandel

MS, Statistics. University of Colorado Denver 2024

- Awarded “Outstanding Masters’ Student” in Fall 2024
- Project Title: Custom Loss Functions in Fuel Moisture Modeling

BS, Statistics. University of British Columbia 2016

Publications

Hirschi, J., Mandel, J., Kochanski, A. (2026). *Transfer Learning for Dead Fuel Moisture Prediction Using Time-Warping Recurrent Neural Networks*. arXiv Preprint arXiv:2605.08379. <https://arxiv.org/abs/2605.08379>

Hirschi, J., Mandel, J., Hilburn, K., Farguell, A. (2026). *A Recurrent Neural Network for Forecasting Dead Fuel Moisture Content with Inputs from Numerical Weather Models*. Fire, 9(1), 26. <https://doi.org/10.3390/fire9010026>

Hirschi, J. (2025). *Custom Loss Functions in Fuel Moisture Modeling*. arXiv Preprint arXiv:2501.10401.

Mandel, J., **Hirschi, J.**, Kochanski, A. K., Farguell, A., Haley, J., Malia, D. V., Shaddy, B., Oberai, A. A., Hilburn, K. A. (2023). *Building a fuel moisture model for the coupled fire-atmosphere model WRF-SFIRE from data: From Kalman filters to recurrent neural networks*. In *Proceedings of the Seminar on Numerical Analysis (SNA'23)* (pp. 52–55). Institute of Geonics of the Czech Academy of Sciences.

Research Experience

Graduate Research Assistant, University of Colorado Denver 2022–Present

- Developing machine learning models of fuel moisture content and data acquisition pipelines as part of NASA wildfire research grants under Dr. Jan Mandel.

Graduate Research Assistant, University of Colorado Denver 2022–2023

- Assisted in developing interactive open-access materials as part of the Open Educational Resources grant project under Dr. Troy Butler.
- Developed and maintained reproducible instructional content and notebooks for applied regression and differential equations courses, supporting interactive, code-driven learning (Repositories: Applied Regression Analysis, ODEs).

Employment

Statistician, Greenfield Advisors

2016–2021

- Prepared expert witness testimony for litigation related to environmental contamination and mortgage-backed securities fraud.

Conference Presentations

Hirschi, J., Mandel, J., Hilburn, K. A., Farguell. *Transfer learning with recurrent neural networks of fuel moisture content: Extending from 10h fuels to 1h and 100h*. American Geophysical Union Annual Meeting (2025 - Poster)

Hirschi, J., *Transfer learning with recurrent neural networks of fuel moisture content: Extending from 10h fuels to 1h and 100h*. Workshop in Numerical Wildfires and AI Weather, Cargèse, Corsica, France (2025).
<https://forefireapi.github.io/cargese2025/program/>

Hirschi, J., Mandel, J., et al., *A Recurrent Neural Network Model of Fuel Moisture Content with HRRR Weather Model Data and VIIRS Satellite Data as Predictors*, AMS 15th Fire and Forest Meteorology Symposium, Denver, CO, (2025)

Hirschi, J., Mandel, J., Hilburn, K. A., Farguell, A., Kochanski, A. *Machine Learning Models of Fuel Moisture Content with Custom Loss Functions*. American Geophysical Union Annual Meeting (2024 - Poster)

Mandel, J., **Hirschi, J.**, Kochanski, A., Farguell, A., Mallia, D. V., Shaddy, B., Oberai, A. A., Hilburn, K. A., & Haley, J. *A Fuel Moisture Model for WRF-SFIRE from HRRR and RAWS Data by a Physics-Initialized Recurrent Neural Network*. 104th American Meteorological Society Annual Meeting (2024). <https://ams.confex.com/ams/104ANNUAL/meetingapp.cgi/Paper/436992>

Mandel, J., **Hirschi, J.**, Kochanski, A., Farguell Caus, A., Haley, J., Mallia, D. V., Shaddy, B., Oberai, A. A., & Hilburn, K. A. *Machine Learning for Fuel Moisture in Time and Space, Dead or Live*. 35th Conference on Agricultural and Forest Meteorology/14th Fire and Forest Meteorology Symposium/Sixth Conference on Atmospheric Biogeosciences. American Meteorological Society, (2023). <https://ams.confex.com/ams/35AF14F6BG/meetingapp.cgi/Paper/424323>

Hirschi, J. *Fuel Moisture Data Acquisition & Spatiotemporal Cross Validation*. Denver, CO. SIAM Front Range Student Conference (2023). <https://github.com/jh-206/FRAMSC-2024—FMDA-Data-and-CV-Methods>

Hirschi, J. *A Bayesian Autoregressive Model of Fuel Moisture Content in Colorado*. Denver, CO. Data 2 Policy Symposium (2023). <https://digital.auraria.edu/work/ns/276d0874-220d-4292-8c4f-e9e8431e6a3b>

- Awarded *Best in Data Analysis*, Spring 2023 Symposium.

Hirschi, J. *The Impact of Wildfire Activity on Stream Temperature in Colorado*. Denver, CO. Data 2 Policy Symposium (2021). <https://digital.auraria.edu/work/ns/c1fd77bb-fea0-4c04-b0f6-3c043540ab8f>

- Awarded *Best in Data Analysis* Spring 2021 Symposium.

Public Software Repositories

- ML methods for fuel moisture data collection and forecasting, Open Wildfire Modeling Group: GitHub: [ml_fmda](#)
- ML methods for transfer learning with fuel moisture data and prediction tools: GitHub: [fmc_transfer](#)

Teaching Experience

Graduate Teaching Assistant, University of Colorado Denver

Fall 2021

- Math 1110, College Algebra. Recitation section instructor.

Certifications

CITI Responsible Conduct of Research
QGIS Workshop, CU Denver FAST Lab

2022
2022